

Tema 6

Métodos de Series de Fourier

Funciones periódicas y series trigonométricas

$$\frac{d^2 x}{dt^2} + \omega_0^2 = f(t) \rightarrow f(t) \text{ es una función armónica simple} \rightarrow \text{Método de coeficientes indeterminados}$$
$$x_p(t) = \frac{A}{\omega_0^2 - \omega^2} \cos(\omega t) \rightarrow x_p(t) = \sum_{n=1}^N \frac{A_n}{\omega_0^2 - \omega_n^2} \cos \omega_n t$$
$$x_p = a \cos \omega t$$

Teorema: Se dice que una función es periódica si:

$$f(t+p) = f(t)$$

Series de Fourier: $f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos(mt) + b_m \sin(mt))$ $\rightarrow$ Si $f(t)$ es periódica en 2π

$$\int_{-n}^n \cos(mt) \cos(nt) dt = \begin{cases} 0 & \text{si } m \neq n \\ n & \text{si } m = n \end{cases}$$

$$\int_{-n}^n \sin(mt) \sin(nt) dt = \begin{cases} 0 & \text{si } m \neq n \\ n & \text{si } m = n \end{cases}$$

$$\int_{-n}^n \cos(mt) \sin(nt) dt = 0$$

$$\int f(t) dt = \frac{a_0}{2} \int_{-n}^n dt + \sum_{m=1}^{\infty} \int_{-n}^n \cos mt dt + \sum_{m=1}^{\infty} b_m \sin(mt) = n a_0 \rightarrow$$

$$a_0 = \frac{1}{n} \int_{-n}^n f(t) dt, \quad a_n = \frac{1}{n} \int_{-n}^n f(t) \cos(nt) dt, \quad b_n = \frac{1}{n} \int_{-n}^n f(t) \sin(nt) dt$$

$$\text{Ej: } f(t) = \begin{cases} -1 & \text{si } -\pi < t < 0 \\ 1 & \text{si } 0 < t < \pi \\ 0 & \text{si } t = -\pi, 0, \pi \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt = \int_{-\pi}^0 (-1) dt + \int_0^{\pi} (1) dt = \frac{1}{\pi} \left([-t]_{-\pi}^0 + [t]_0^{\pi} \right) = \frac{1}{\pi} (-\pi + \pi) = 0$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt = \int_{-\pi}^0 -\cos(nt) dt + \int_0^{\pi} \cos(nt) dt = \frac{1}{\pi} \left(\left[-\frac{1}{n} \sin(nt) \right]_{-\pi}^0 + \left[\frac{1}{n} \sin(nt) \right]_0^{\pi} \right)$$

$$a_n = \frac{1}{\pi} \left(\frac{1}{n} \sin(-\pi n) + \frac{1}{n} \sin(\pi n) \right) = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^0 -\sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} \sin(nt) dt = \frac{1}{\pi} \left(\left[\frac{1}{n} \cos(nt) \right]_{-\pi}^0 + \left[-\frac{1}{n} \cos(nt) \right]_0^{\pi} \right) =$$

$$\frac{1}{\pi} \left(\left[\frac{1}{n} - \frac{1}{n} \cos(n\pi) \right] + \left[-\frac{1}{n} \cos(n\pi) + \frac{1}{n} \right] \right) \Rightarrow \frac{1}{\pi} \left(\frac{1}{n} (1 - \cos(n\pi) - \cos(n\pi) + 1) \right) = \frac{2}{\pi n} (1 - \cos(n\pi)) = \frac{2}{\pi n} (1 - (-1)^n)$$

$\cos(n\pi) = \cos(-n\pi) = (-1)^n$

$$b_n = \begin{cases} \frac{4}{n\pi} & \text{si } n \text{ impar} \\ 0 & \text{si } n \text{ par} \end{cases}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt)$$

Para n impares: $f(t) = \sum_{\substack{n=1 \\ \text{impares}}}^{\infty} \frac{4}{n\pi} \sin nt = \frac{4}{\pi} \sum_{\substack{n=1 \\ \text{impares}}}^{\infty} \frac{\sin(nt)}{n}$

Formulas de Integracion Utiles:

$$\int u \cos u du = \cos u + u \sin u + C$$

$$\int u \sin u du = \sin u - u \cos u + C$$

$$\int u^n \cos u du = u^n \sin u - n \int u^{n-1} \sin u du$$

$$\int u^n \sin u du = -u^n \cos u + n \int u^{n-1} \cos u du$$

$$E_3: \begin{cases} 0 & \text{si } -\pi < t \leq 0 \\ t & \text{si } 0 \leq t < \pi \\ \pi/2 & \text{si } t = \pm \pi \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} t \, dt = \frac{1}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} t \cos nt \, dt = \frac{t}{n} \sin(nt) - \int \frac{1}{n} \sin(nt) \, dt = \left[\frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_0^{\pi} = \frac{1}{n^2} \left[n t \sin(nt) + \cos(nt) \right]_0^{\pi}$$

$$h = t \quad v = \frac{1}{n} \sin nt$$

$$dh = dt \quad dv = \cos nt$$

$$a_n = \frac{1}{n^2} (\cos n\pi - 1) = \frac{1}{n^2} [(t-1)^n - 1] \quad a_n = \begin{cases} -\frac{2}{n^2} & \text{si } n \text{ es impar} \\ 0 & \text{si } n \text{ par} \end{cases}$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} t \sin nt \, dt = \frac{1}{\pi} \left[-\frac{t}{n} \cos(nt) + \int \frac{1}{n} \cos(nt) \, dt \right]_0^{\pi} = \frac{1}{n^2} \left[\sin(nt) - n \cos(nt) \right]_0^{\pi} = \frac{1}{n^2} (n \cos(n\pi) - n)$$

$$h = t \quad dh = dt \\ v = \frac{1}{n} \cos(nt) \quad dv = -\sin(nt)$$

$$b_n = \frac{1}{n\pi} (\cos(n\pi) - 1) = \frac{1}{n\pi} ((-1)^n - 1) \rightarrow b_n = \begin{cases} -\frac{2}{n\pi} & \text{si } n \text{ impar} \\ 0 & \text{si } n \text{ par} \end{cases}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt) = \frac{\pi}{4} - \frac{2}{\pi} \sum_{n \text{ impar}} \frac{\cos(nt)}{n^2} + \frac{\sin(nt)}{n}$$

Serie de Fourier General y Convergencia

Considerese $f(t)$ una función continua por tramos de período $2L$ definida en toda t . Entonces la serie de Fourier de $f(t)$ es la serie:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L}$$

donde los coeficientes de Fourier:

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \left(\frac{n\pi t}{L} \right) dt$$

donde L es el semiperíodo.

$$\text{Ej: } f(t) = \begin{cases} -1 & \text{si } -2 < t < 0 \\ 1 & \text{si } 0 < t < 2 \end{cases}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt = \frac{1}{2} \int_{-2}^2 f(t) dt = \frac{1}{2} \left[\int_{-2}^0 -1 dt + \int_0^2 1 dt \right] = \frac{1}{2} \left[[-t]_{-2}^0 + [t]_0^2 \right] = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt = \frac{1}{2} \int_{-2}^2 f(t) \cos \left(\frac{n\pi t}{2} \right) dt = \frac{1}{2} \left(\int_{-2}^0 -\cos \left(\frac{n\pi t}{2} \right) dt + \int_0^2 \cos \left(\frac{n\pi t}{2} \right) dt \right) = 0$$

$$b_n = \frac{1}{2} \int_{-2}^2 f(t) \sin \left(\frac{n\pi t}{2} \right) dt = \frac{1}{2} \left(\int_{-2}^0 -\sin \left(\frac{n\pi t}{2} \right) dt + \int_0^2 \sin \left(\frac{n\pi t}{2} \right) dt \right) = \frac{1}{2} \left[\left[\frac{2}{n\pi} \cos \left(\frac{n\pi t}{2} \right) \right]_{-2}^0 + \left[-\frac{2}{n\pi} \cos \left(\frac{n\pi t}{2} \right) \right]_0^2 \right]$$

$$b_n = \frac{1}{2} \left[\frac{2}{n\pi} - \frac{2}{n\pi} \cos(n\pi) - \frac{2}{n\pi} \cos(n\pi) + \frac{2}{n\pi} \right] = \frac{2}{n\pi} (1 - \cos(n\pi)) = \frac{2}{n\pi} (1 - (-1)^n) \quad b_n = \begin{cases} \frac{4}{n\pi} & \text{si } n \text{ impar} \\ 0 & \text{si } n \text{ par} \end{cases}$$

$$f(t) = \frac{4}{\pi} \sum_{n \text{ impar}} \frac{\sin \left(\frac{n\pi t}{2} \right)}{n}$$

Teorema: Supóngase que la función periódica f es suave por tramos. Entonces su serie de Fourier dada converge:

- 1) al valor $f(t)$ en cada punto donde f es continua
- 2) al valor $\frac{1}{2} [f(t+) + f(t-)]$ en cada punto donde f es discontinua.

Soluciones de ecuaciones diferenciales por serie de Fourier

1. Se extiende la definición de la función $f(t)$ al intervalo $-L < t < 0$
2. Después a toda la recta real mediante condiciones de periodicidad $f(t+2L) = f(t)$

$$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L}$$

$$\begin{aligned} \text{Ej: } x'' + 4x &= 4t & f(t) &= 4t \text{ para } 0 < t < 1 \\ x(0) &= x(1) = 0 \end{aligned}$$

Extensión impar de periodo 2